Sal^{2}RN: A Spatial–Spectral Salient Reinforcement Network for Hyperspectral and LiDAR Data Fusion Classification

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***Abstract*— Hyperspectral image (HSI) and light detection and ranging (LiDAR) data fusion have been widely employed in HSI classification to promote interpreting performance. In the existing deep learning methods based on spatial–spectral features, the features extracted from different layers are treated fairly in the learning process. In reality, features extracted from the continuous layers contribute differentially to the final classification, such as large tracts of woodland and agriculture typically count on shallow contour features, whereas deep semantic spectral features have meaningful constraints for small entities like vehicles. Furthermore, the majority of existing classification algorithms employ a patch input scheme, which has a high probability to introduce pixels of different categories at the boundary. To acquire more accurate classification results, we propose a spatial–spectral saliency reinforcement network (Sal^{2}RN) in this article. In spatial dimension, a novel cross-layer interaction module (CIM) is presented to adaptively alter the significance of features between various layers and integrate these diversified features. Moreover, a customized center spectrum correction module (CSCM) integrates neighborhood information and adaptively modifies the center spectrum to reduce intraclass variance and further improve the classification accuracy of the network. Finally, a statistically based feature weighted combination module is constructed to effectively fuse spatial, spectral, and LiDAR features. Compared with traditional and advanced classification methods, the Sal^{2}RN achieves the state-of-the-art classification performance on three open benchmark datasets.**

***Index Terms*— Fusion classification, hyperspectral image (HSI), light detection and ranging (LiDAR), spatial–spectral feature.**

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I. Introduction

HYPERSPECTRAL sensors capture image cubes in dozens to hundreds of narrow bands, and the hyperspectral images (HSIs) realize the combination of spectral and spatial information effectively. HSIs with high spectral resolution has been widely investigated in agriculture, biomedicine and mineral exploitation [1], [2], [3], [4], etc. In the field of academic research, HSI classification aims to assign unique label for each pixel according to the wealthy discriminative signatures, which is a significant technology to perceive and interpret hyperspectral imagery.

In the early studies of HSI classification, many traditional machine learning (ML)-based methods come to the fore, including benchmarking support vector machine (SVM) using kernel principle [5], logistic regression (LR) [6], random forest [7], and so on. Although these ML algorithms have achieved superior classification performance, they all rely on manual feature creation and shallow features for learning and discrimination, which cannot distinguish deep and complex features precisely in HSIs [8].

In recent decades, due to the powerful feature representation ability of deep learning (DL) [9], it has flourished and employed in various vision applications [10]. DL methods are capable of learning sufficient features and fitting parameters from original data, which have been proved to be more competitive than traditional methods in many vision tasks [11]. The representative DL structures include recurrent neural networks (RNN) [12], long short-term memory (LSTM) [13], convolutional neural network (CNN) [14], etc. Particularly, CNN has become one of the most robust approaches in remote sensing image processing owing to its unique receptive field structure and parameters sharing of convolution kernel [14]. Hu et al. [15] utilizes CNN to classify hyperspectral data and achieves encouraging classification accuracy. However,

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the feature extraction of HSI only in the spectral dimension, dubbed 1-D convolution (1D-CNN) which is applied to operate along the spectral dimension of the image without giving full play to the spatial context information [16]. Aiming at this issue, HybridSN [17] combines the advantages of 2-D convolution (2D-CNN) and 3-D convolution (3D-CNN), and fully extracts spectral-spatial features. Subsequently, an increasing number of CNN methods for hyperspectral classification based on spatial–spectral features extraction has been proposed to enhance the classification accuracy. For instance, [18] designs the involution operator applying to the spatial–spectral features of HSIs. Recently, transformer structures have arose vigorously in the field of natural language processing due to the aggrandizing amount of available data [19]. The transformer structure makes full use of the self-attention mechanism to extract features [19], which overcome the distance problem in sequence data. At the same time, many researches inherit the advantages of transformer to the field of image processing and design vision transformer (ViT) networks based on transformer idea [20]. Meanwhile, transformer structure network for HSI processing also emerged. Hong et al. [21] design the SpectralFormer network to apply the ViT structure to HSI classification for the first time. SpectralFormer designs collaborative patch input mode to extract spectral features in the neighborhood. Moreover, the cross-layer feature fusion mode is designed to fuse the data of transformer from different layers and achieves excellent classification performance. However, the transformer structure consumes more computing resources because it requires a large amount of self-attention calculation. To solve this problem, research [22] uses the unique shift windows (Swin) transformer structure. This variant of transformer structure with hierarchical design not only introduces local information, but also greatly saves computation. In addition, based on the characteristics of the image itself, [23] adopts random mask operation to reduce the computation load without damaging the accuracy.

With the escalating demands for precision and the advancement of sensor technology, several researches have devoted their attention to fuse data from various sensors. For instance, light detection and ranging (LiDAR) data assisted by photoelectric detecting systems contain precise elevation information [24], [25], and the performance of the network in feature extraction and classification can be considerably enhanced by the combination of HSI and LiDAR data [26], [27]. Typical networks include two-branch network [28], hierarchical random walk network (HRWN) [29], FusAtNet [30], Couple CNN [31], etc. These methods successfully interacted with certain significant semantic features of these two types of data during the feature extraction stage. However, due to the insufficient annotated data and the fusion data with high dimension, these types of networks face the problem of over-fitting.

In general, a straightforward guideline to surmount the aforementioned deficiency is expanding the sample data volume which can promote the ability of the model to learn more abundant features. The typical CNN-based pixel-pair features (PPFs) [32] select two pixels randomly from the whole training set, deduce the labels of each pixel pair with subtle rules, and generate more training samples. In the meantime, the employment of multiscale connections in different layers can make the learned features more representative and distinct through concating with shallow features progressively, which alleviates the over-fitting to some extent. In view of this, Hyperspectral Pyramidal ResNet [33] converges shallow features with deep features through pyramid bottleneck residuals to improve the robustness. DenseNet [34] interacts each layer with other layers through features reusing, which effectively strengthens the feature representational ability, while decreases the parameters. These aforementioned networks extract features interdependencies through skip connection, but they ignore the different contributions of features from different layers to perception. In detail, the buildings with small areas and discrete distribution resort to deeper semantic context features, while large and continuous ground objects such as farmland and forest dependent on more shallow shapes and contour features.

More recently, many remarkable algorithms adopted sharding input schemes to capture highly delicate spatial–spectral features. Since the center pixel in ground object data frequently resembles to the surrounding pixels, the patch input mode is undertaken to capture extra identification features around the center pixel [35], [36], [37]. Specifically, Zhang et al. [38] proposed an unsupervised patch-to-patch CNN (PToP) network for spatial–spectral collaborative classification. However, the utilization of patch input with constant size still has dilemmas that do not guarantee the adjacent pixels in the same category, especially near edges [39]. Besides, the spectral discrepancies between different categories may stymie the classifier. Toward this end, several researches exploit multiscale inputs to circumvent the interference of fixed size in the network to tackle this problem. Multiscale input can effectively deal with the features of different sizes, so as to classify different objects. In general, a smaller granularity can restore the details of the image, while a sparser granularity can advance the features of the global context [40]. For instance, Yang et al. [41] propounded a cascaded dilated convolutional network (CDCN), employing dilated convolution to enlarge the receptive field for achieving multiscale features in spatial dimensions. The aforesaid algorithm alters the receptive field and draw upon patches of various sizes to perceive refined border details, but there are still boundary spectrums that differ from the center pixel, which pose a deficiency to the capabilities of the perceptron.

To conquer the above challenges, we propose an innovative spatial–spectral salient reinforcement network (Sal^2RN) to achieve superior classification performance. In Sal^2RN, we present a feature weighted combination module utilizing compatibility fusion strategy (CFS) after feature extractor, which diminishes the influence of simple superposition of features on robustness. Moreover, a novel cross-layer interaction module (CIM) is designed to aggregate the features of multilayer networks. In comparison to standard ResNet and DenseNet, Sal^2RN can leverage CIM to better weight the features between layers. This module not only decreases the risk of over-fitting, but also enlarges the ability of network to

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adapt to a variety of ground objects. Furthermore, the center spectrum correction module (CSCM) in Sal²RN leverages global average pooling (GAP) in all dimensions of HSI data to effectively mitigate the multiclass pixel interference of input patches. The modified spectrums comprise the average spectral information of all pixels in the patch, which significantly ameliorates the network's robustness for pixels of multiple categories in a patch. The main contributions can be summarized as follows.

1) A novel HSI classification network Sal²RN is proffered. For detailed ground object classification, the proposed network can extract spatial, spectral, and elevation properties from HSI and LiDAR data and effectively integrate them through a feature weighted combination module.
2) The CIM which as a central component is embedded in the spatial feature extraction stage can gratifyingly highlight the contributions of diverse layers of the network to classification and make feature fusion more effective.
3) The proposed CSCM modulates the spectrum of the central pixel point by exploiting the neighbor pixel information given by patch, reducing intraclass variance and alleviating the challenge of the same object with varied spectra.
4) The Sal²RN has obtained remarkable classification results on three hyperspectral fusion classification datasets. Specifically, the proposed network has achieved 93.45%, 98.80%, and 92.71% overall accuracy (OA) on Houston, TRENTO, and MUUFL Gulfport dataset, respectively.

The following related works will be introduced in Section II. In Section III, each part of the network will be described in detail, and the experimental results and analysis will be presented in Section IV. Finally, the conclusion of this article is in Section V.

II. Related Works

A. Hyperspectral Data Fusion Method

Hyperspectral multisource data fusion is mainly divided into three categories: pixel-level fusion, feature-level fusion and decision-level fusion. In the proposed network, pixel-level fusion and feature-level fusion are applied for classification. Pixel-level image fusion is the lowest level fusion method, which merges registered multisource remote sensing images into a single fusion image pixel by pixel. The advantage of pixel-level fusion is that it is less time consuming, easier to implement, and the resulting image contains more original information [42]. In HSI, high-dimensional data not only bring a lot of information, but also have a lot of redundant information, so it is necessary to reduce the dimension of data before pixel fusion. Zhao et al. [43] used principal component analysis (PCA) combined with the IHS transform and weighted fusion method to complete dimensional-reduction pixel fusion of the image while retaining the main information. However, pixel-level fusion is greatly affected by noise and misregistration [44]. In addition, the Gram–Schmidt fusion method has high image fidelity and is also an efficient image fusion method [45], and the process of the Gram–Schmidt algorithm is as follows:

$$\mathbf{GS}_T(i,j) = (\mathbf{O}_T(i,j) - \mu_T) - \sum_{l-1}^{T-1} \phi(\mathbf{O}_T, \mathbf{GS}_l) * \mathbf{GS}_l(i,j) \tag{1}$$

where T is the number of transformed bands, $\mathbf{O}$ is the original band, and μ_T is the mean value of channel T

$$\mu_T = \frac{\sum_{j=1}^{C}\sum_{i=1}^{R}\mathbf{O}_T(i,j)}{C * R} \tag{2}$$

where C and R represent the number of pixels in each row and column of the image, respectively, and $\phi(\mathbf{O}_T, \mathbf{GS}_l)$ is given by

$$\phi(\mathbf{O}_T, \mathbf{GS}_l) = \left[\frac{\sigma(\mathbf{O}_T, \mathbf{GS}_l)}{\sigma(\mathbf{GS}_l, \mathbf{GS}_l)}\right]. \tag{3}$$

And the σ is given by

$$\sigma = \sqrt{\frac{\sum_{j=1}^{C}\sum_{i=1}^{R}(\mathbf{O}_T(i,j) - \mu_T)}{C * R}}. \tag{4}$$

Because of its superiority, the Gram–Schmidt image fusion method is adapted in the data input stage to fuse multisource remote sensing data.

Feature-level fusion is an intermediate level of integration, which is to fuse and classify various features extracted from the input image, including size, length, edge, texture, and shape. The basic idea is to use the fusion algorithm based on region block. That is, the image is divided into regions which are very informative, and the features of each region are combined. Typical networks such as Deeplab series network [46] and Unet series [47] network utilize feature fusion and reinforcement of different depths to carry out semantic segmentation. In [48], a multiscale feature fusion layers (MFLs) were designed using feature-level fusion to fuse semantic features and detail features, and significant performance improvement was achieved. Zuo et al. [49] designed a deep multicascade (DMC) structure to fuse features and place them in the feature extraction network to selectively fuse useful information from different modes. As the most commonly employed data fusion method, feature-level fusion has a wide range of applications. At present, many researches also tend to carry out feature-level fusion after extracting multisource features, which can effectively improve the accuracy of the final downstream task. In this work, we make full use of the advantages of pixel-level fusion and decision-level fusion to fuse and classify HSIs and LiDAR, and achieve excellent performance.

B. Attention Mechanism in HSI Classification

Attention mechanism is an important part of human vision and has become a research hotspot in the past few years [50]. The attention mechanism pays attention to the relationship between samples. When extracting features, individual features are not only related to themselves, but also related to other elements. Using a weight matrix to describe the correlation between samples can make the input features have better global information. On the whole, the attention mechanism

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enhances the ability of network to perceive effective information and enables the model to focus on the pivotal parts of the features [51]. Many studies have adopted the attention mechanism in their own networks and achieved significant improvements; typical network models such as squeeze and excitation network (SENet) [52]. In this network, a feature recalibration strategy is proposed for the first time, which obtains the degree of importance for each feature channel through learning, so that the training model can achieve better results. Attention mechanism also plays a crucial role in the field of HSI classification. For instance, spectral-spatial attention network (SSAN) [53] designs spectral attention RNN branch and spatial attention CNN branch to jointly extract spatial–spectral features, and achieves excellent classification performance. To better learn the representative information in the data, Haut et al. integrated ResNet with the attention mechanism to effectively identify the important spectral-spatial features [54].

C. Hyperspectral and LiDAR Fusion Classification

In recent years, many researches combine HSI with LiDAR images for classification. Typical networks are two-branch CNN [28] and HRWN [29]. The former designs a unique network which has two branches to extract HSI features and LiDAR image features, respectively. Specifically, 2-D convolution structure and 1-D convolution are applied in HSI features extraction to capture spectral and spatial features simultaneously. And a cascade block is utilized in LiDAR feature extraction for feature reusing. Finally, the features are stacked and classified in the classifier. Based on the two-branch network, HRWN applies the pixel affinity approach to the LiDAR branch. In addition, a hierarchical random walk module is designed in the classifier to fuse the features of different sources and obtain more significant classification results. Moreover, the FusAtNet [30] applies the attention mechanism for landcover classification. In the feature extraction part, the self-attention method is adopted to gather HSI and LiDAR features, and then this network links multimodal data with cross-attention. Couple CNN [31] also achieves satisfactory results. This network adopts the strategy of sharing parameters when extracting features from different sources, greatly reducing the number of operation parameters, and designs a decision fusion module to better adapt to the features of multisource data. Different from the above methods, the proposed method adopts an exceptional feature fusion strategy and achieves better classification performance.

III. Proposed Method

This section introduces the proposed Sal^{2}RN model in detail. First, the overall structure of the network is introduced. Then, the details of each part of the Sal^{2}RN are illustrated.

A. Spatial–Spectral Saliency Reinforcement Network

Pixel-level and feature-level fusion of multisource images are different methods of fusion analysis, and they, respectively, focus on the source images and the features extracted from DL networks. Based on the above principles, Sal^{2}RN simultaneously utilizes two fusion methods to fully fuse multisource images. The proposed network also designs a unique feature extractor to extract features from fused HSIs, and an additional branch to extract the features of the LiDAR image.

Fig. 1 depicts the overall structure of Sal^{2}RN which consists of three parts: input part of the pixel level fusion of HSI and LiDAR data, the feature extraction module, and the features fusion classification module with CFS. For the first fusion of the multi-source (MS) data, we adapt Gramm–Schmidt method to align HSI data with LiDAR data elevation information. After initial fusion, we generate input data containing both spectral information and spatial elevation information and divide the fusion data and LiDAR data into patches $\mathbf{X}_i^f \in \mathbb{R}^{r\times r\times N}$ and $\mathbf{X}_i^l \in \mathbb{R}^{r\times r\times 1}$, respectively, where $r \times r$ represents the size of input patch, and N is the channel numbers of each HSI patches. And these patches serve as the input for the next module.

For the feature extraction module, we utilize a three-branch network to obtain the spectral, spatial context and elevation information.

In the first branch of the spatial, the fused patches $\mathbf{X}_i^f$ are fed into a four-layer dense block structure with dilated convolution and the CIM which is designed to strengthen the connection of features between contiguous layers. Then, after the full connection layer, we can get the output features of spatial branch

$$\mathbf{F}_{\text{spa}} = f_{\text{CIM}}(\mathbf{F}_n, \mathbf{F}_{n-1}) \tag{5}$$

where $\mathbf{F}_n$ and $\mathbf{F}_{n-1}$ denote the result of the final dense block structure and that of the previous layer, respectively. And f_{CIM} indicates the CIM operation. For the flow of spectral feature extraction, we design a CSCM that utilizes factors from adjacent pixels. The input of this branch is fusion patch $\mathbf{X}_i^f$ and the center pixel point of each patch. Then the spectral output can be concatenated as $\mathbf{F}_{\text{spec}}$. Due to the error accumulation in the process of Gramm–Schmidt fusion, there may be partial loss of LiDAR data, so we add an additional LiDAR branch to extract its features $\mathbf{F}_{\text{lid}}$ separately and complement the initial data at the end.

After the feature extraction of three branches, the salient feature reinforcement module is designed. The three types of features are integrated and classified in this module.

B. Dense Block and CIM

Traditional attention approach [55], or residual structure, works by concatenating the results of the previous layer and the current layer. The indiscriminate penetration of past information not only adds features with weak correlation to the current layer, but also fails to highlight the multisource data with strong correlation. Based on the above issues, we propose a novel CIM, the detailed structure of which is displayed in Fig. 2. Dense structure with dilated convolution is employed in shallow feature extraction network, and the dilation ratios are set to 1, 2, 5, 1 []. The dilated convolution can expand the receptive field during feature extraction, which is suitable for extracting spatial context information. Then, the CIM concatenates the output of the last layer $\mathbf{F}_n$ and the features of the previous layer $\mathbf{F}_{n-1}$ as input. $\mathbf{F}_n$ through a 2-D convolution

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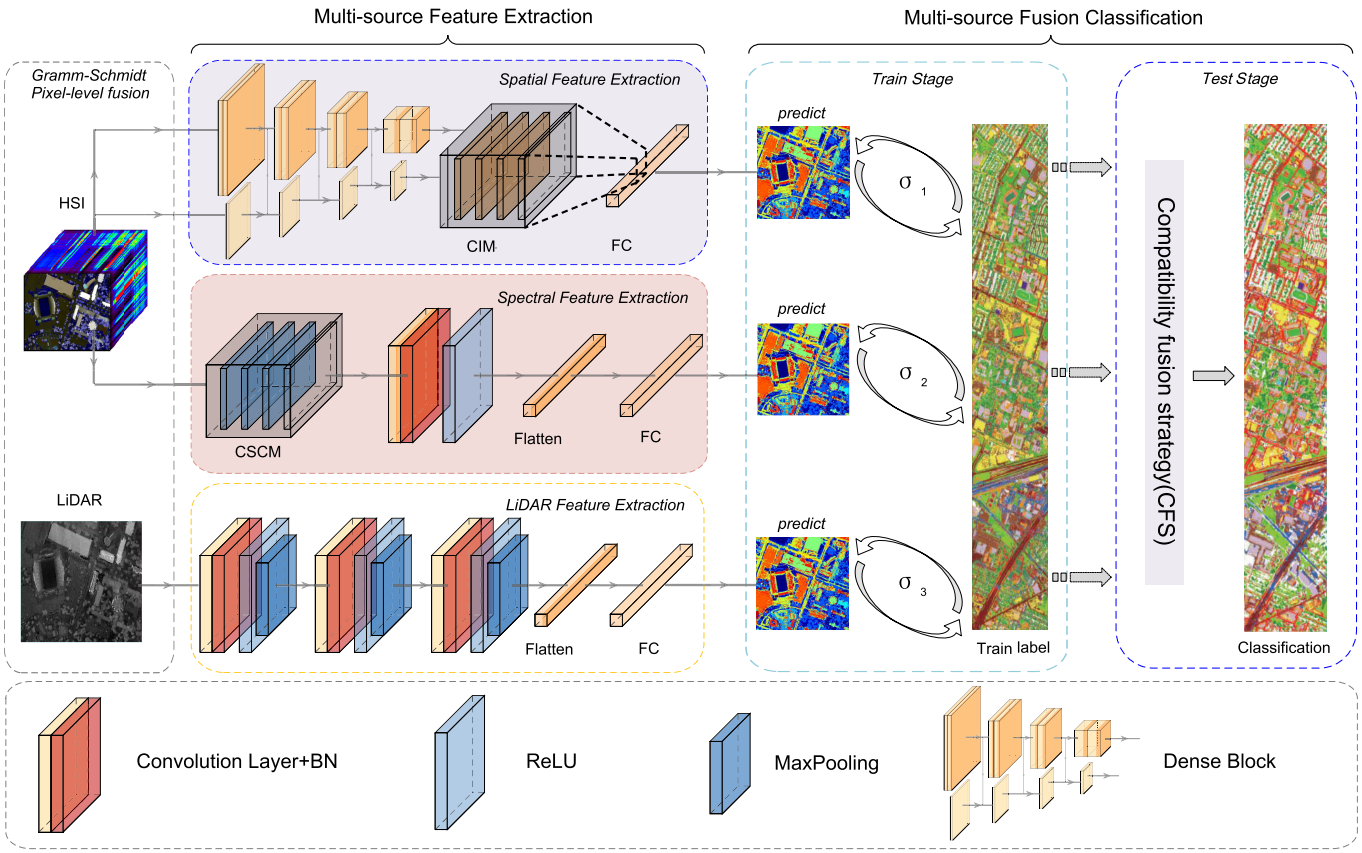


Fig. 1. Framework of the proposed Sal^{2}RN. It consists of two parts: multisource feature extraction network and multisource fusion classification network. Feature extraction is divided into three branches, in which CIM and CSCM are the CIM and CSCM, respectively. The dense block is a four-layer dense connection structure, which is applied for initial spatial feature extraction.

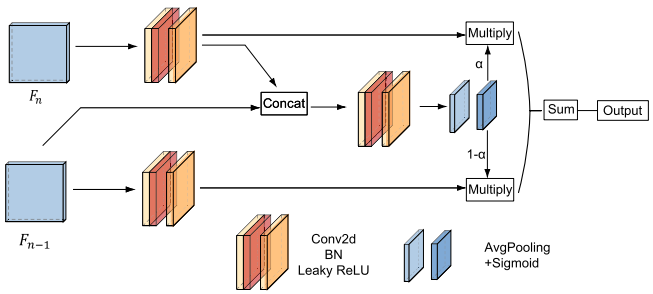


Fig. 2. Structure of CIM.

layer with a kernel size of 3×3, and merged with $\mathbf{F}_{n-1}$. The fusion features pass through a convolution layer with a kernel size of 3×3, then pass through a GAP, and the activation function is the standard Sigmoid function. The entire module is defined as

$$\mathbf{F}'_n = f(\mathbf{W} \cdot \mathbf{F}_n + \mathbf{b}) \tag{6}$$

$$\mathbf{Q} = \mathbf{Avg}\left(f\left(\mathbf{F}'_n \parallel \mathbf{F}_{n-1}\right)\right) \tag{7}$$

where $\mathbf{W}$ is the weights of the layer, $\mathbf{b}$ is the bias, and $\parallel$ denotes the concatenating operation of the features.

The features that are joined together through a GAP, and then we utilize the Sigmoid function as the activation function to generate the weighted probability vector of the joint features

$$\boldsymbol{\alpha} = \frac{\exp(\mathbf{Q})}{\exp(\mathbf{Q}) + 1} \tag{8}$$

where the $\boldsymbol{\alpha}$ represents a weight vector obtained by the above operations. Then we dot $\boldsymbol{\alpha}$ and $1 - \boldsymbol{\alpha}$ with $\mathbf{F}'_n$ and $\mathbf{F}'_{n-1}$

$$\mathbf{F}_{\text{spa}} = \mathbf{Concat}\left(\boldsymbol{\alpha} \cdot \mathbf{F}'_n + (1 - \boldsymbol{\alpha}) \cdot \mathbf{F}'_{n-1}\right). \tag{9}$$

$\mathbf{F}_{\text{spa}}$ is the final output of the CIM, as well as the output of the entire spatial branch. By combining the two inputs and performing convolution operation, the correlation between inputs at different layers can be effectively learned, and this correlation is denoted by $\boldsymbol{\alpha}$. Modify the two inputs $\mathbf{F}'_n$ and $\mathbf{F}'_{n-1}$ with different weights, respectively, can effectively emphasize the weightiness of different layers to the final classification.

C. CSCM and LiDAR Branch

We design a spectral branch to correct the spectral information of hyperspectral data. Considering the spectral correlation between each pixel and the pixels within the surrounding patch for the following reasons.

First, in HSIs, there exists the phenomenon of homogeneity, that is, pixels representing the same category may have different spectral information. In addition, as shown in Fig. 3, hyperspectral data basically belong to the same category in

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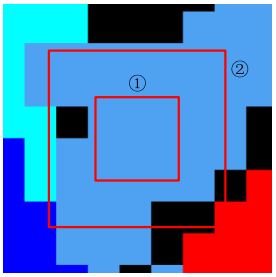


Fig. 3. Illustration of different patch sizes, various colors represent different ground objects.

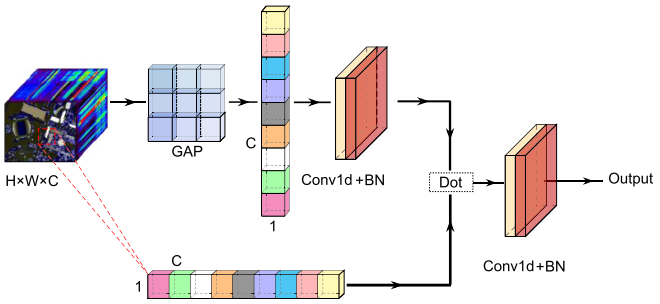


Fig. 4. Structure of CSCM.

the range around each pixel, but when patch is adapted as the input, other categories of pixels are likely to be included in the boundary. These extra pixels have a very different spectral distribution and can also interfere with the classifier. Therefore, the spectral information of all pixels in the input patch is used to correct the pixels of the central category, which can effectively reduce the influence of the above phenomenon on classification results. Given an HSI dataset $\mathbf{X} \in \mathbb{R}^{C\times N}$, where N is the number of spectral bands and C represents the pixel number of the dataset. Then, dividing $\mathbf{X}$ into patches $\mathbf{X}_i \in \mathbb{R}^{r\times r\times N}$ with the size of $r \times r$, and the center pixel of each patch $\mathbf{X}_i^{\text{cen}}$ is utilized as the category of the whole patch. To fuse the spectral information of the surrounding pixels, both $\mathbf{X}_i^{\text{cen}}$ and $\mathbf{X}_i$ are used as the input of the spectral branch. The detailed structure of CSCM is shown in Fig. 4, a 2-D GAP is carried out along the spectral channel of the input patches, and $r \times r$ pixels of each patch are averaged

$$\mathbf{F}_i = \frac{1}{r^2}\sum_{j=1}^{r}\sum_{k=1}^{r}\mathbf{X}_{jk} \tag{10}$$

where $\mathbf{F}_i = [\mathbf{F}_1, \ldots, \mathbf{F}_N] \in \mathbb{R}^{1\times N}$ represents the bands dimension features about each patch, and $\mathbf{X}_{jk}$ denotes the element of the jth row and kth column in $\mathbf{X}_i$. After obtaining the mean value feature, we employ a 1-D convolution layer and rectified linear unit (ReLU) layer to learn the feature correlation between each channel. Then, the generated mean weight vector $\mathbf{F}^* \in \mathbb{R}^{\mathbf{1}\times\mathbf{N}}$ is dotted with the central pixel of the patch $\mathbf{X}_i^{\text{cen}}$, and this process can be written as follows:

$$\mathbf{S} = \mathbf{X}_i^{\text{cen}} \otimes \mathbf{F}_i^* \tag{11}$$

where $\otimes$ denotes the dot product operation, and $\mathbf{S}$ represents the central spectrum after patch modification. Then a 1-D convolution layer and a fully connected layer are utilized for feature extraction of the spectral

$$\mathbf{F}_{\text{spe}} = \sigma(f(\mathbf{W}_s\mathbf{S} + \mathbf{b}_s)) \tag{12}$$

where $\mathbf{F}_{\text{spe}}$ is the final output of this channel. f and σ represent 1-D feature extraction operation and full connection layer along the bands, and $\mathbf{W}_s$ and $\mathbf{b}_s$ denotes the weights and the bias, respectively.

In addition, an additional LiDAR feature extraction branch is designed to compensate for the loss of elevation information in pixel fusion. Since the LiDAR image is a single channel image, a deep network with too many layers will be unable to learn elevation features, but rather focus on deep semantic information. We utilize a three-layer shallow network to learn elevation features, as shown in Fig. 1. The specific parameters of the branch network structure are as follows: the kernel size of all convolutional layers is set to 3×3, and each convolutional layer is followed by a batch normalization (BN) layer, a ReLU layer, and a maximum pooling layer of size 2. Then a full connection layer is adapted to extract LiDAR image features $\mathbf{F}_{\text{lid}}$.

D. Salient Features Reinforcement Module and CFS

Through the previous three-branch feature extraction network, we obtain the features extracted from the input image from different angles. In the traditional feature fusion methods, many methods simply spliced various features together without distinguishing the contribution degree of these features to the final classification result. In this article, we utilized CFS to fuse the features. Specifically, different features are assigned different weights, which can not only emphasize the importance of different features in classifier, but also can avoid the feature extraction network learning that has adverse effect on classification results. The details of this section are shown in Fig. 1, and the CFS can be described by Algorithm 1.

Algorithm 1 CFS

Input: Output of spatial feature extraction network, $\mathbf{F}_{spa}$; Output of spectral feature extraction network, $\mathbf{F}_{spe}$; Output of LiDAR feature extraction network, $\mathbf{F}_{lid}$;

1: Initialization: $\varphi_{spa}^n, \varphi_{spe}^n, \varphi_{lid}^n$, EPOCH, $n \leftarrow 0$
2: Using the learned tripartite features for training model prediction, separately. And the prediction maps $pred_{spa}^n, pred_{spe}^n, pred_{lid}^n$ is obtained.
3: The prediction map is compared with the real training map pixel by pixel, and the number of correctly classified pixels is counted, and the probability of correct classification $\varphi_{spa}^n, \varphi_{spe}^n, \varphi_{lid}^n$ is obtained.
4: **repeat**
5: $n \leftarrow n + 1$
6: Update $pred_{spa}^n, pred_{spe}^n, pred_{lid}^n$ based on training model.
7: Update $\varphi_{spa}^n, \varphi_{spe}^n, \varphi_{lid}^n$ based on step 3
8: **until** The number of epochs reaches a preset value

Output: Features fusion output $\mathbf{F}_{out}$ via (17)

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First of all, feature maps generated by spectral, spatial, and LiDAR branches are applied to predict the classification results, respectively. Then, we compare the classification result with the real training ground truth pixel by pixel and count the number of correctly classified pixels

$$\text{count} = \begin{cases} \text{count} + 1, & \text{if } \mathbf{pred}_i = \mathbf{gt}_i \\ \text{count} + 0, & \text{otherwise} \end{cases} \tag{13}$$

where $\mathbf{pred_i}$ is the classification result generated by prediction, and $\mathbf{gt_i}$ is the real value of the corresponding position. Then calculate the proportion, and the formula is as follows:

$$\varphi_{\text{spa}} = \text{count}_{\text{spa}}/gt \tag{14}$$

$$\varphi_{\text{spe}} = \text{count}_{\text{spe}}/gt \tag{15}$$

$$\varphi_{\text{lid}} = \text{count}_{\text{lid}}/gt \tag{16}$$

where *count* represents the number of correctly classified pixels, and *gt* represents the number of all pixels to be classified. And the rearranged fusion feature output is expressed as

$$\mathbf{F}_{\text{out}} = \varphi_{\text{spa}} \cdot \mathbf{F}_{\text{spa}} + \varphi_{\text{spe}} \cdot \mathbf{F}_{\text{spe}} + \varphi_{\text{lid}} \cdot \mathbf{F}_{\text{lid}}. \tag{17}$$

Then we adapt the rearranged features to predict the model and get the final classification results.

IV. Experimental Results and Analysis

In this section, we utilize three hyperspectral remote sensing data and the corresponding LiDAR image to test the effectiveness of the proposed network. To verify the convincingness of the network, we compare it with many classical hyperspectral classification networks, such as 3-D convolution neural network (3-D CNN) and CNN-PPF network based on DL. Meanwhile, we adopt a DL network for HSI classification employing spectral-spatial features, dubbed two-branch CNN network with dual branches for simultaneous processing of spectral and spatial information. To compare the performance of the multisource fusion model, we also adapt two most advanced fusion and classification networks using HSIs and LiDAR images, namely Couple CNN and HRWN. Besides, ViT and SpectralFormer based on the latest transformer architecture are used to demonstrate the excellence of the proposed network.

For easy distinction, algorithms using only hyperspectral data are labeled "H" and algorithms utilizing both hyperspectral and LiDAR data are marked "H+L". All the algorithms mentioned above are well trained and the codes are implemented with Python3.6. And the Sal²RN is built using PyTorch framework, equipped with graphics processing unit (GPU) NVIDIA GTX2080Ti.

A. Experimental Data

For multisource fusion classification, three main datasets are included.

1) *Houston Data:* This dataset was introduced during the GRSS Data Fusion Contest in 2013 and includes a HSI and a LiDAR derived digital surface model (DSM) which were collected at the University of Houston campus and in adjacent urban areas. The HSI contains 144 spectral bands, distributed in 380–1050 nm region, and has a spatial resolution of 2.5 m. The size of the image is 349 × 1905, including 15,029 ground-truth samples distributed on 15 specific ground object categories, and the number of pixels in the training set and test set is 2,832 and 12,197. Detailed description of dataset categories is shown in Table I.
2) *TRENTO Data:* This dataset was taken in a rural area of the Italian city of Trento. The HSIs contain 63 spectral bands and are distributed in the region of 420–990 nm, with a spatial resolution of 1 m. Including 30,214 ground-truth samples distributed on six specific ground object categories, and the number of pixels in the training set and test set is 819 and 29 395. Detailed description of dataset categories is shown in Table II.
3) *MUUFL Gulfport Data:* The dataset was taken at the University of Mississippi's Bay Park campus. The HSIs contain 64 spectral bands and are distributed in the region of 380–1050 nm, with a spatial resolution of 1 m. Including 53 732 ground-truth samples distributed on 11 specific ground object categories, In the training set, 100 samples were randomly selected from each class, and the rest sample points were adapted as test sets. Detailed description of dataset categories is shown in Table III.

TABLE I
CATEGORIES AND SAMPLE NUMBERS FOR THE HOUSTON DATA

Class		Samples Number	
No.	Name	Train	Test
1	Health grass	198	1053
2	Stressed grass	190	1064
3	Synthetic grass	192	505
4	Trees	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking lot1	192	1041
13	Parking lot2	184	285
14	Tennis court	181	247
15	Running track	187	473
Total		2832	12197

TABLE II
CATEGORIES AND SAMPLE NUMBERS FOR THE TRENTO DATA

Class		Samples Number	
No.	Name	Train	Test
1	Trees	129	3905
2	Buildings	125	2778
3	Ground	105	374
4	Woods	154	8969
5	Vineyard	184	10317
6	Roads	122	3525
Total		819	29395

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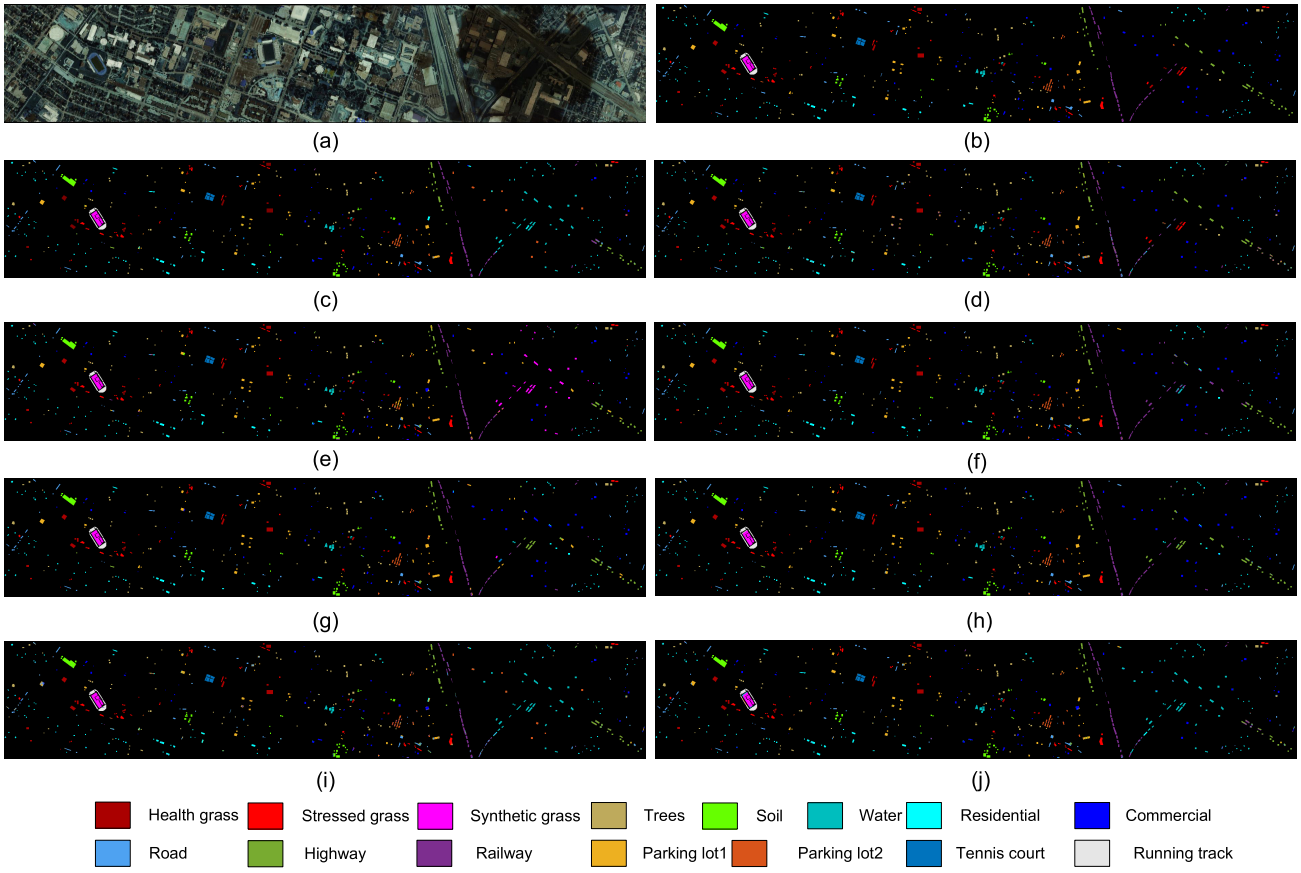


Fig. 5. Maps of classification results via different methods on the Houston dataset. (a) False-color image. (b) Ground truth. (c) CNN-PPF. (d) 3D-CNN. (e) Two-branch CNN. (f) Couple CNN. (g) HRWN. (h) ViT. (i) SpectralFormer. (j) Sal^{2}RN.

TABLE III
CATEGORIES AND SAMPLE NUMBERS FOR THE MUUFL GULFPORT DATA

Class		Samples Number	
No.	Name	Train	Test
1	Trees	100	23146
2	Mostly grass	100	4170
3	Ground surface	100	6782
4	Dirt	100	1762
5	Road	100	6587
6	Water	100	366
7	Building shadow	100	2133
8	Building	100	6140
9	Sidewalk	100	1285
10	Yellow curb	100	83
11	Cloth panels	100	169
Total		1100	52632

B. Classification Performance and Analysis

To evaluate the superior performance of the proposed network, several experiments are carried out on the dataset, and using the measures of average accuracy (AA), OA, and kappa coefficient (*ka*). Tables IV through VI show the performance of the above methods on the Houston, TRENTO, and MUUFL Gulfport datasets. For the accuracy of the experiment, each method is repeated five times, and the final result is the average of the output results of each experiment. Meanwhile, the tables in bold and underlined represent the best and second-best classification results for each feature category, respectively.

Table IV depicts the performance of each method on the Houston dataset. It can be observed from the classification performance that the proposed network has advantages in OA, AA, and Kappa coefficients. Specifically, compared with the second-best algorithm, AA index improved by 1.66%, OA index increased by 1.99%, and Kappa coefficient increased by 2.15%. At the same time, due to the fusion of LiDAR data, HRWN and Couple CNN have better classification results than the network using hyperspectral data alone, which proves that the auxiliary elevation information in LiDAR data can effectively improve the classification accuracy. In addition, the Houston data have a discrete distribution characteristic, and the proposed method still achieves excellent classification results.

Table V delineates the classification results of different algorithms on the MUUFL Gulfport dataset. It can be concluded from the table that the proposed method achieves the best effect in all three indicators. Concretely, AA is 93.45%, OA is 92.57%, and Kappa coefficient is 0.9193. Meanwhile, when the number of category 6, 10, and 11 samples is very small, the classification results can be more than 90%. In addition, the boundary of the dataset is relatively complex,

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TABLE IV
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON HOUSTON DATASET

No.	CNN-PPF(H)	3D CNN(H)	Two-Branch(H+L)	Couple CNN(H+L)	HRWN(H+L)	ViT(H)	SpectralFormer(H)	Ours(H+L)
1	83.00%	**98.30%**	83.10%	82.43%	85.31%	82.72%	81.86%	82.91%
2	84.12%	98.68%	84.87%	84.87%	83.79%	80.45%	**100.00%**	85.15%
3	**100.00%**	99.53%	**100.00%**	99.80%	99.05%	99.60%	95.25%	93.27%
4	88.54%	94.30%	92.14%	92.06%	92.30%	92.42%	**96.12%**	91.67%
5	**100.00%**	98.82%	97.73%	**100.00%**	**100.00%**	97.73%	99.53%	**100.00%**
6	97.20%	89.45%	68.53%	97.20%	97.28%	95.80%	94.41%	**100.00%**
7	83.40%	79.89%	87.33%	**92.91%**	89.33%	74.44%	83.12%	90.86%
8	46.25%	82.41%	70.75%	96.01%	93.74%	42.55%	76.73%	**96.30%**
9	84.04%	79.36%	84.51%	84.99%	88.66%	65.25%	79.32%	**93.96%**
10	56.37%	84.96%	62.64%	67.47%	**86.17%**	50.77%	78.86%	83.59%
11	80.08%	72.32%	76.47%	98.57%	92.75%	71.44%	88.71%	**98.86%**
12	87.42%	80.55%	91.26%	96.15%	96.47%	56.00%	87.32%	**97.21%**
13	82.81%	89.73%	8.12%	84.91%	**91.93%**	64.21%	72.63%	88.07%
14	**100.00%**	99.74%	**100.00%**	**100.00%**	**100.00%**	**100.00%**	**100.00%**	**100.00%**
15	98.94%	99.34%	98.93%	99.58%	**100.00%**	98.52%	**100.00%**	**100.00%**
AA(%)	84.81%	89.82%	82.70%	91.79%	90.47%	78.13%	88.91%	**93.45%**
OA(%)	81.69%	88.54%	80.42%	90.58%	89.67%	74.36%	88.01%	**92.57%**
Kappa	0.803	0.8761	0.8124	0.8978	0.8828	0.7243	0.8699	**0.9193**

TABLE V
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON MUUFL GULFPORT DATASET

No.	CNN-PPF(H)	3D CNN(H)	Two-Branch(H+L)	Couple CNN(H+L)	HRWN(H+L)	ViT(H)	SpectralFormer(H)	Ours(H+L)
1	88.34%	82.27%	86.88%	94.66%	**95.20%**	87.62%	88.83%	94.45%
2	81.49%	81.04%	77.19%	**85.08%**	84.72%	81.29%	66.62%	81.45%
3	77.25%	67.58%	83.57%	77.04%	72.93%	59.99%	71.73%	**87.44%**
4	93.57%	85.64%	95.71%	97.45%	**98.20%**	82.73%	88.47%	96.18%
5	88.90%	83.00%	**94.55%**	86.18%	85.35%	80.12%	84.21%	93.98%
6	99.18%	91.34%	61.20%	**100.00%**	**100.00%**	84.97%	92.62%	**100.00%**
7	90.06%	86.12%	83.54%	**95.59%**	94.33%	79.79%	86.45%	94.24%
8	81.12%	71.94%	94.79%	96.16%	92.82%	82.88%	83.37%	**96.67%**
9	72.14%	71.39%	63.97%	74.86%	64.36%	71.67%	74.24%	**85.34%**
10	80.72%	91.73%	54.22%	96.39%	85.54%	**97.59%**	89.16%	90.91%
11	97.63%	95.89%	94.08%	**99.41%**	97.63%	95.86%	96.45%	99.16%
AA(%)	86.40%	82.54%	80.87%	90.44%	88.28%	82.24%	83.83%	**92.71%**
OA(%)	85.53%	79.32%	86.95%	91.17%	89.32%	81.23%	83.24%	**92.59%**
Kappa	0.8122	0.7364	0.8301	0.8745	0.8589	0.7564	0.7818	**0.9019**

and the visualization results show that our network is far better than other comparison algorithms in defining the boundary.

The performance of each method in the TRENTO dataset is shown in Table VI. Although Couple CNN and HRWN methods based on LiDAR and HSI fusion images have excellent performance, the proposed Sal²RN still has the highest classification accuracy. Compared with the second-best algorithm, AA, OA, and Kappa coefficients increased by 2.27%, 1.42%, and 2.74%, respectively. As shown in Fig. 7(b), the distribution of TRENTO dataset is relatively concentrated, that is, ground objects of the same type are mostly distributed in the same area. The Sal²RN can effectively extract spatial context information and supplement the edge information of different ground objects with LiDAR data, thus achieving excellent classification results.

The visualized results of different classification methods are shown in Figs. 5–7. In addition, for the convenience of observation, the pseudo-color map and ground-truth map of each type of dataset are also drawn. From the visualization results, we can observe that the Sal²RN method has the greatest similarity with ground truth in the classification result maps. Specifically, in the Houston dataset, due to the lack of spatial context information support, some classification methods will have more classification errors in the same category of ground objects. At the same time, the distribution of all kinds of ground objects is scattered, and the boundary information is fuzzy. In the algorithm without LiDAR data fusion, the result of boundary classification is not satisfactory. In general, the proposed method has better classification results for fewer pixels in discrete datasets. In TRENTO dataset, due to its uniform distribution, the CSCM method proposed by us can better balance the spectrum of each category of interior objects, so the classification results are more complete and there are few errors in the classification of some pixels in the category. The same is true for GUUFL Gulfport dataset, at the boundary and small sample points, we have fewer errors in classification compared with other algorithms.

C. Ablation Study

In our proposed network, CIM and CSCM are applied in feature extraction module to improve the performance of the network. Therefore, to verify the effectiveness of CIM

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TABLE VI
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON TRENTO DATASET

No.	CNN-PPF(H)	3D CNN(H)	Two-Branch(H+L)	Couple CNN(H+L)	HRWN(H+L)	ViT(H)	SpectralFormer(H)	Ours(H+L)
1	97.13%	99.22%	91.45%	99.13%	89.29%	87.35%	96.08%	**99.51%**
2	92.12%	90.50%	**97.83%**	95.43%	91.22%	81.21%	95.86%	97.48%
3	98.93%	97.90%	92.48%	**99.73%**	83.72%	96.79%	95.99%	96.52%
4	99.10%	97.05%	98.31%	99.51%	98.08%	97.42%	97.99%	**99.99%**
5	96.71%	94.09%	99.51%	98.84%	**100.00%**	74.66%	95.25%	**100.00%**
6	68.32%	79.48%	83.08%	**93.25%**	87.27%	69.95%	57.76%	91.76%
AA(%)	94.14%	93.04%	96.19%	98.19%	95.55%	84.57%	92.37%	**98.80%**
OA(%)	92.05%	93.86%	93.84%	97.24%	91.60%	83.70%	89.82%	**97.54%**
Kappa	0.9216	0.9183	0.9419	0.9758	0.9403	0.7844	0.8982	**0.9839**

TABLE VII
RESULTS OF CLASSIFICATION IN ABLATION STUDIES FOR CIM AND CSCM OF THE NETWORK WE PROPOSED

Dataset	Indicator	Basic Feature	Basic Feature +CIM	Basic Feature +CSCM	Basic Feature +CIM+CSCM
Houston	AA(%)	87.75%	89.20%	88.44%	**93.45%**
	OA(%)	86.68%	88.15%	88.69%	**92.57%**
	kappa	0.8556	0.8716	0.8756	**0.9193**
TRENTO	AA(%)	93.67%	95.86%	95.69%	**98.80%**
	OA(%)	95.17%	97.85%	97.04%	**97.54%**
	kappa	0.9554	0.9712	0.9634	**0.9839**
MUUFL	AA(%)	88.16%	91.93%	91.94%	**92.71%**
	OA(%)	88.05%	90.42%	91.74%	**92.59%**
	kappa	0.8634	0.8936	0.8909	**0.9019**

TABLE VIII
RESULTS OF CLASSIFICATION IN ABLATION STUDIES FOR CFS

Dataset	Indicator	Feature	Feature +CFS
Houston	AA(%)	87.09%	**93.45%**
	OA(%)	87.50%	**92.57%**
	kappa	0.8622	**0.9193**
TRENTO	AA(%)	98.11%	**98.80%**
	OA(%)	96.73%	**97.54%**
	kappa	0.9746	**0.9839**
MUUFL	AA(%)	90.20%	**92.71%**
	OA(%)	90.77%	**92.59%**
	kappa	0.881	**0.9019**

and CSCM, some ablation experiments are conducted in this section. Specifically, the suggested feature extraction structure is broken down into three sections: basic feature, CIM, and CSCM. We test the final classification results on the three datasets using the same experimental settings. The experimental results of the final arrangement are shown in Table VII. As can be inferred from the results in the table, the proposed method achieves the best classification results on all datasets. Specifically, in the Houston dataset, OA decreased by 4.42% and 3.88%, respectively, in the absence of CIM and CSCM. In TRENTO dataset, OA decreased by 0.39% and 1.5% in the absence of CIM and CSCM, respectively. In the MUUFL Gulfport dataset, OA decreased by 2.17% and 0.85% in the absence of CIM and CSCM, respectively. This indicates that combining CIM and CSCM in the feature extraction step can improve the capture of spatial and spectral features more effectively. In addition, the table also shows the classification results of the network in the case that neither CIM nor CSCM is utilized, which also confirms the help of the proposed two modules for feature acquisition when they are implemented alone.

In this section, we also verify the designed multisource feature fusion module, that is, the module which adopts CFS. Specifically, we replace the original feature fusion module with simple superposition of features while keeping the network feature extraction module unchanged. The same experimental settings are employed to verify the final classification performance on three different datasets, and the detailed classification performance is shown in Table VIII. It can be observed from the table that the salient feature reinforcement module has obtained the best classification results on the three basic datasets. Classification accuracy improved by 5.07% on the Houston dataset, 0.81% on the TRENTO dataset, and 1.82% on the MUUFL Gulfport dataset. In the Houston dataset, a large improvement has been obtained, and the preliminary analysis is due to the different distribution of data. Compared with the gt map of the other two datasets, the distribution of the Houston dataset is scattered and the categories are disorganized, so CFS can better process the features of different sources and achieve a more significant improvement in classification accuracy.

D. Discussion on Input Patch Size

In this section, we explore the influence of input patches of different sizes on the final classification results of the network. In particular, we adjust the input of the patch size of 3×3, 5×5, 7×7, 9×9, 11×11, 13×13, and 15×15, respectively. The network classification performance of the three datasets is tested without changing other parameters. The specific classification results are plotted in Fig. 8. And we can observe from the figure that when we adopt smaller input sizes, such as 3, 5, and 7, the classification results are not ideal. The analysis reason is that when the size is small, the context information contained in the image is also very limited, and the effect of CSCM in the spectral branch is not satisfactory. On the contrary, when we adapt a larger size, such as 13 and 15, there will be too many pixels of other categories in the same patch, which will affect the classification result.

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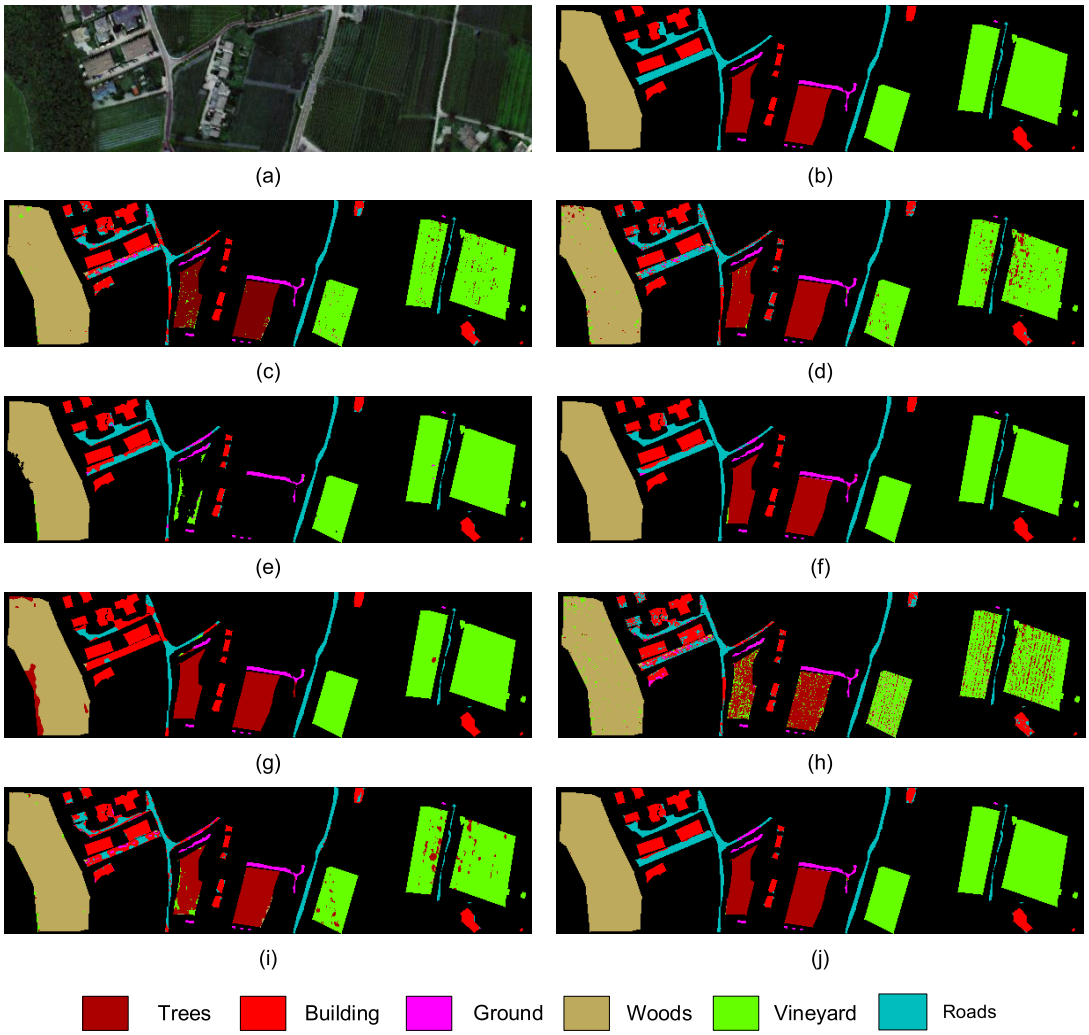


Fig. 6. Maps of classification results via different methods on the TRENTO dataset. (a) False-color image. (b) Ground truth. (c) CNN-PPF. (d) 3D-CNN. (e) Two-branch CNN. (f) Couple CNN. (g) HRWN. (h) ViT. (i) SpectralFormer. (j) Sal^{2}RN.

TABLE IX
TRAINING AND TESTING TIMES, NUMBER OF PARAMETERS AND FLOPS UNDER THE TRENTO DATASET

	CNN-PPF	3D CNN	Two-Branch	Couple CNN	HRWN	ViT	SpectralFormer	Ours
Training Times(s)	1202.24	279.56	117.43	71.57	64.71	1749.62	1985.46	253.15
Testing Times(s)	2.41	2.55	15.24	1.3	20.86	5.37	6.21	2.82
Params(M)	0.12	0.46	0.14	0.3	0.16	2.43	0.52	0.9
FLOPS(M)	10.47	1.28	3.74	1.74	3.24	30.43	33.49	11.59

However, due to the correction effect of CSCM on the spectrum, the classification result of larger patch is better than that of small patch. Uniquely, patch15 has achieved more significant classification results on TRENTO dataset. The analysis is due to the distribution form of TRENTO dataset, which contains a large area of forest and farmland, so larger patch input will have more excellent spatial context information and better classification results. However, for Houston and MUUFL Gulfport datasets with complex ground object categories, patch size of 11 has better classification effect. To ensure the unity of the network structure, the input patch size is selected as 11 according to the classification results of the three datasets.

E. Computational Cost

In addition, for different algorithms, we calculate the training and testing times, the number of parameters and floating-point operations per second (FLOPS) under the TRENTO dataset.

We design the network extracts features from different sources and integrates them for classification, and achieve excellent classification performance. As can be seen from

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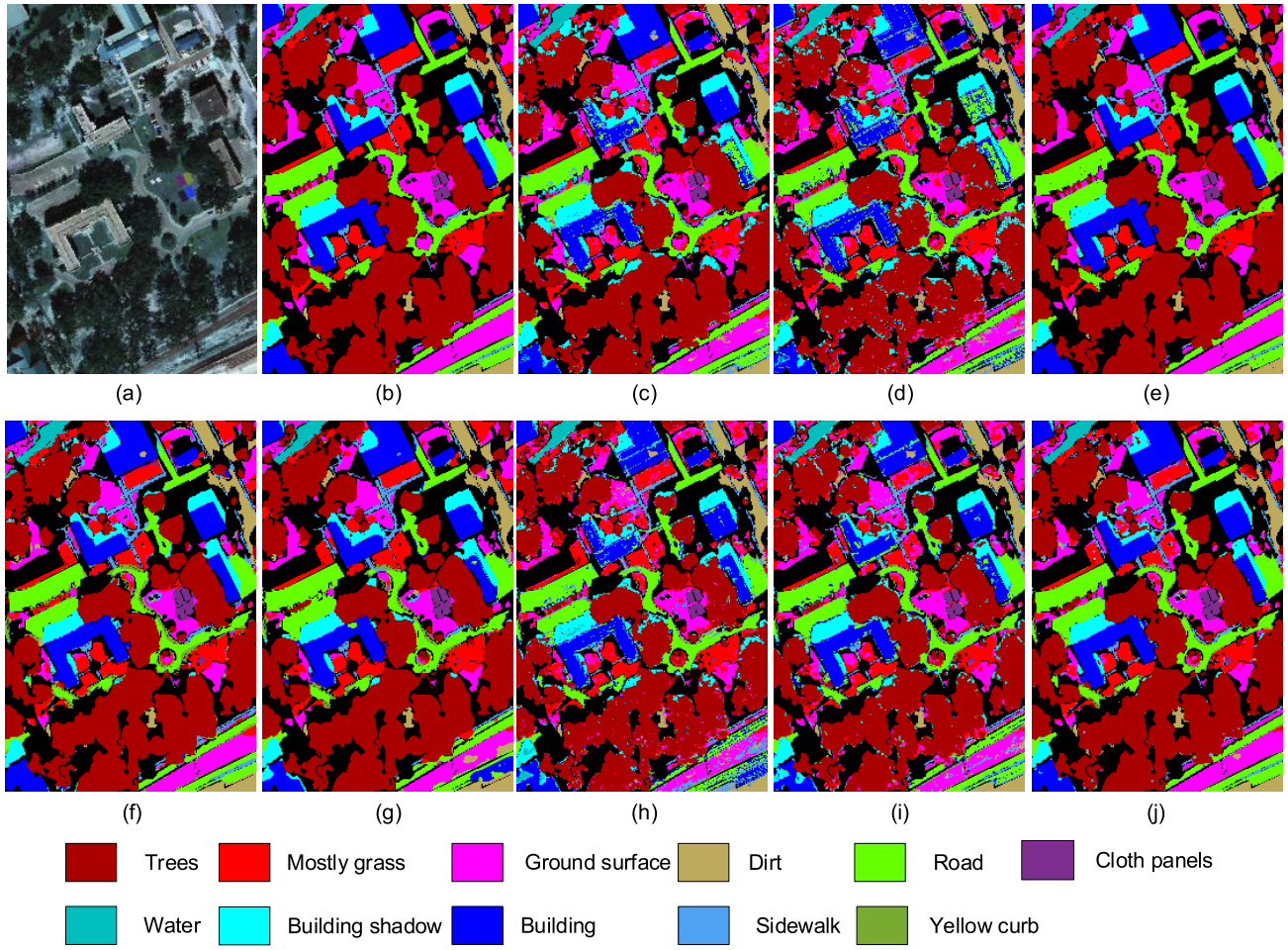


Fig. 7. Maps of classification results via different methods on the MUUFL Gulfport dataset. (a) False-color image. (b) Ground truth. (c) CNN-PPF. (d) 3D-CNN. (e) Two-branch CNN. (f) Couple CNN. (g) HRWN. (h) ViT. (i) SpectralFormer. (j) Sal^{2}RN.

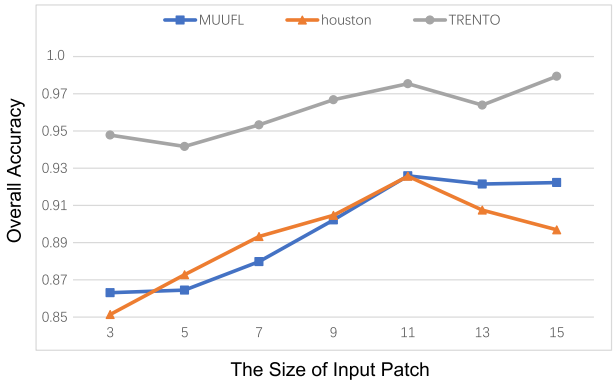


Fig. 8. OA results obtained on three different datasets with different patch sizes.

Table IX, our network is far less than ViT and SpectralFormer networks with transformer structure in terms of the number of parameters and operation time. Due to the Dense structure and the adoption of a multibranch network, the FLOPS of our network may be slightly higher than the others. However, considering the classification results in Table VI the computational cost is acceptable compared with the final performance improvement.

V. Conclusion

In this article, we propose a novel DL network based on multisource fusion and CFS for HSI classification. First of all, to alleviate the redundant superposition of features in the traditional attention algorithm, CIM is designed in the feature extraction stage to enhance the fusion of features of different layers, so as to reduce the impact of redundant features on the classifier. Second, to reduce the influence of surrounding pixels on the central pixels in the patch input, CSCM is adapted during spectral feature extraction, and the remaining pixels are utilized to adjust the center pixels to reduce interference from the phenomena that the same object with different spectrums. Finally, we design a feature fusion module based on CFS to deal with multisource features. The experimental results show that the proposed Sal^{2}RN has a better performance than other HSI classification algorithms.

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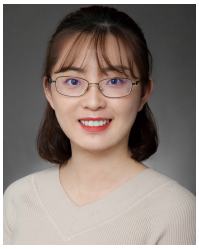
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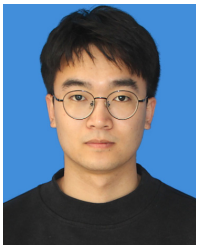
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